Exam Date & Time: 10-May-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Data Analytics with Python [NPT-3201]

Marks: 50 **Duration: 180 mins.**

PART-A

Answer all the questions.

1) Explain different levels of data and discuss different levels of data measurement in detail with appropriate examples. (4)

A)

B) Compare the following-

- Sample Variance vs. Population variance.
- Sample standard deviation vs. Population standard deviation. (3)

C) When are you going to use mean, median and mode to measure central tendency? Justify your answer with suitable examples. (3)

2) Illustrate different types of probability events and discuss them briefly. (5)

A)

B) Shown here are the raw values matrix and corresponding probability for the results of a national survey of 200 executives who were asked to identify the geographic locale of their company and their company's industry type and the executives were only allowed to select one locale and one industry type.

Industry Type	Geographic Location			
	Northeast D	Southeast E	Midwest F	West G
Finance A	24	10	8	14
Manufacturing B	30	16	22	12
Communication C	28	18	12	16

(5)

a) Compute the probability that the respondent is from the Midwest (F)?
b) Compute the probability that the respondent is from the communication industry (C) or from the Northeast (D)?

3) The image contains data representing the study hours per day by four different students (A, B, C, D) on three different days (Monday, Tuesday, and Wednesday). The questions posed are:

A)

Days	A	B	C	D
Monday	2	3	4	5
Tuesday	4	4	6	6
Wednesday	6	5	8	8

(5)

Examine whether the study hours of the different students are the same.
Examine whether the study hours on different days are the same.

B) What is the probability distribution? Explain with a suitable example. Also Write assumptions of Binomial distribution. (5)

4) Explain how Maximum Likelihood Estimation (MLE) is utilized in the parameter estimation for a Logistic Regression model. Discuss the significance of the Receiver Operating Characteristic (ROC) curve in evaluating the performance of such a model. Additionally, outline the steps involved in building a Regression Analysis Model and how it differs from Logistic Regression. Your answer should (5)

A)

include examples of scenarios where each model would be appropriately applied

B) Construct a decision tree using the provided dataset and then formulate the prediction rules for the accompanying test data.

Patient ID#	Sore Throat	Fever	Swollen Glands	Congestion	Headache	Diagnosis
1	Yes	Yes	Yes	Yes	Yes	Strep throat
2	No	No	No	Yes	Yes	Allergy
3	Yes	Yes	No	Yes	No	Cold
4	Yes	No	Yes	No	No	Strep throat
5	No	Yes	No	Yes	No	Cold
6	No	No	No	Yes	No	Allergy
7	No	No	Yes	No	No	Strep throat
8	Yes	No	No	Yes	Yes	Allergy
9	No	Yes	No	Yes	Yes	Cold
10	Yes	Yes	No	Yes	Yes	Cold

(5)

Data Instances with an Unknown Classification

Patient ID#	Sore Throat	Fever	Swollen Glands	Congestion	Headache	Diagnosis
11	No	No	Yes	Yes	Yes	?
12	Yes	Yes	No	No	Yes	?
13	No	No	No	No	Yes	?

5) For the given dataset compute Entropy,Information gain,gini index and list the best splitting attribute.

A)

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	−
4	F	F	4.0	+
5	F	T	7.0	−
6	F	T	3.0	−
7	F	F	8.0	−
8	T	F	7.0	+
9	F	T	5.0	−

(5)

B) The dataset provided contains information about individuals' weekly hours spent on both social media platforms and at the gym. We aim to determine the presence of any clusters within this dataset through the application of Hierarchical Cluster Analysis. Additionally, please generate a dendrogram to visually represent any identified clusters at the conclusion of the analysis.

	Social media	Gym
Alan	2	3
Lisa	5	2
Joe	5	3
Max	1	4
Caro	4	5

(5)

-----End-----